

# Homework 1 - Intro to Java

**Course:** CIS 284 - Computer Programming 2 - Fall 2026

**Instructor:** Dr. Aaron Mininger

**Institution:** Messiah University

## Submission Instructions

**Due Date:** September 1 at 11:59pm

Complete all the work in this folder, then zip the folder and upload to Canvas.

**You must put your name on every file in the submission**

Note: You may not use any AI assistance on this assignment. *Use only Java features discussed in class.*

Make sure that your program compiles, even if it is incorrect you will receive partial credit.

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### 1. LetterGrade.java

Write a program that asks the user for their grade (as an **int** from 1-100)

The program should then print their letter grade (no plus or minus), with

$A \geq 90$ ,  $B \geq 80$ ,  $C \geq 70$ ,  $D \geq 60$ ,  $F < 60$

Example:

```
Please enter your grade from 1-100:
88
You got a B
```

### Part 2B

Modify your program so that it uses a while loop. It should repeat the calculation above, allowing the user to enter additional grades, until they enter a -1.

Output Example:

```
Please enter your grade from 1-100:
88
You got a B
Please enter your grade from 1-100:
15
You got an F
-1
Goodbye!
```

## 2. Quadratic.java

The quadratic formula is used to find the two roots of the equation  $ax^2 + bx + c = 0$

The formula is given as:  $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Write a program **Quadratic.java** that satisfies the following:

1. Takes 3 numbers  $a, b, c$  as *command line arguments*
2. Prints the full equation  $ax^2 + bx + c = 0$ , inserting values  $a, b$ , and  $c$
3. Calculates and prints the two roots

Example Output:

```
java Quadratic.java 1.0 0.0 -1.0
-----
Equation: 1.0x^2 + 0.0x + -1.0 = 0
root 1 = 1.0
root 2 = -1.0
```

Hint: Use **Math.sqrt(x)** to calculate the square root of a number, and **Double.parseDouble(s)** to convert a String argument to a double.

Part 2B.

Add a check at the top of your program that makes sure 3 command line arguments are given. If there are not 3 arguments, then print an error message and exit.

**Hint:** Use **args.length** to get the number of arguments, and **return;** to exit the program early.

Example:

```
java Quadratic.java 4.5
-----
Error, you need to give 3 arguments
```

Part 2C.

The part inside the square root (  $b^2 - 4ac$  ) is called the *discriminant*. It can tell you how many real solutions there are:

- $disc > 0$  means 2 real solution
- $disc = 0$  means 1 real solution
- $disc < 0$  means 0 real solutions

Modify your program so that it computes the discriminant as an intermediate result, then prints it and the appropriate number of roots and their values.

## Examples:

```
java Quadratic.java 1.0 0.0 -1.0
```

```
-----
```

```
1.0x^2 + 0.0x + -1.0 = 0
```

```
Discriminant: 4.0
```

```
There are 2 real roots
```

```
root 1 = 1.0
```

```
root 2 = -1.0
```

```
java Quadratic.java 1.0 2.0 1.5
```

```
-----
```

```
1.0x^2 + 2.0x + 1.5 = 0
```

```
Discriminant: -2.0
```

```
There are 0 real roots
```